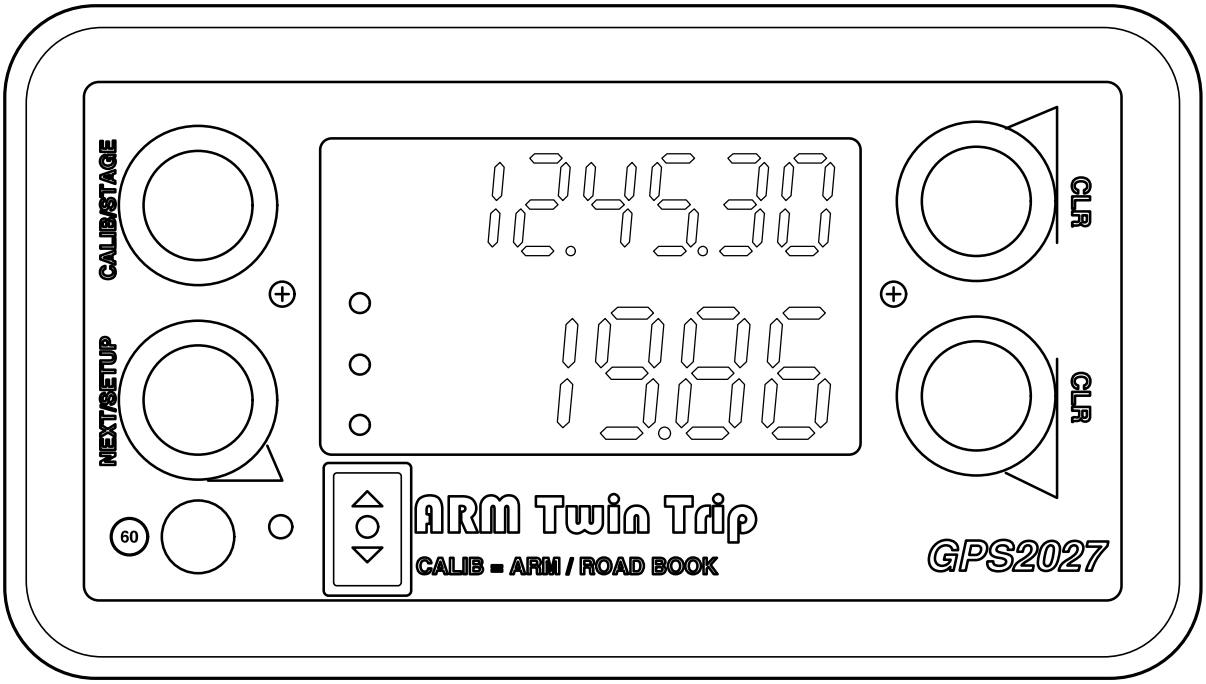
ARM products / GPS Trip Meter

ARM Twin Trip

GPS2027

User Manual

Please read this manual before use and keep it for future reference.



Contents

01	Safety precautions
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CHAPTER 01

Safety precautions

▲ Power supply
Power the unit from **either USB-C (5 V) or DC12V, never both**. Supplying power from both at the same time may damage the unit.

▲ Operating while driving
All operation while driving should be done by the co-driver. The driver must concentrate on driving; mount the driver display where it can be read with minimal eye movement.

▲ GPS start-up time
The first time you use the unit, or after it has been switched off for a long time, it may take several minutes before GPS measurement starts. Switch it on well before the event or drive.

GPS reception
Distance and speed are measured by GPS. Reception may not be possible under a roof or in tunnels. After switching on, check GPS reception (green LED steady) in an open area before use.

Mounting location
Do not use or store the unit where it can get wet or hot (e.g. in direct sunlight on the dashboard).

CHAPTER 02

Parts



No.	Name	Function
1	Red button (CALIB / STAGE)	Short press: stage timing. Long press: distance calibration.
2	Blue button (NEXT / SETUP)	Short press: switches the upper display. Long press: time offset, brightness and driver display settings.
3	Upper CLR	Acts on the upper display (reset etc.). Increases the value while setting.
4	Lower CLR	Resets Trip B. Decreases the value while setting.
5	60 button	Speed alert ON / OFF and setting.
6	Red LED	Speed alert.
7	Direction switch	↑ forward, ○ stop, ↓ reverse.
8	Upper display	Clock, Trip A, speed, final time or stage records.
9	Lower display	Trip B (section distance).
10	Green LED (GPS)	Blinking: searching. Steady: GPS received.
11	White LED (direction)	Off: forward. On: stop. Blinking: reverse.
12	Blue LED	Final time warning (chapter 10).

Right side: from the top, power switch (for DC12V), remote control socket, USB-C power (5 V).

Rear: mounting insert nuts, connectors for the driver display (optional) and the main harness (chapter 14).

CHAPTER 03

Power and GPS

Using USB-C

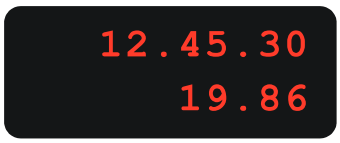
1. Connect 5 V to the USB-C port on the right side. The unit switches on automatically.
2. Right after start-up the display shows “------”.
3. While the green LED blinks, the unit is searching for GPS. When it stays on, GPS is received and the correct time is shown.

Using DC12V

1. Connect DC12V (red) and GND (black) of the main harness on the rear to the vehicle.
2. Press the lower side (|) of the red switch on the right side for ON, the upper side (○) for OFF.
3. The switch is not used when powering from USB-C.



Right after start-up



After GPS reception (clock)

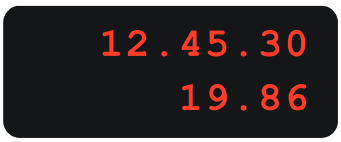
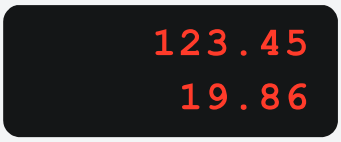
What is kept
Trip A, Trip B, calibration, time offset, brightness, the driver display distance format, speed limit and Ave settings are kept when the power is off. Stage records are not kept.

▲ Caution
Do not connect USB-C and DC12V at the same time.

CHAPTER 04

Switching the upper display

Each short press of the blue button switches the upper display as below (the blue button on the remote does the same). The lower display normally always shows Trip B.

Screen	Upper display	Example
Clock	GPS time (h.mm.ss)	
Trip A	Total distance (km, 10 m steps)	
Speed	“S” followed by the calibrated speed (km/h)	
Final time	Ahead / behind against the Ave (chapter 10)	
Stage records	Stage time. The lower display shows the record number	

After stage records, the display returns to the clock.

CHAPTER 05

Trips (Trip A / Trip B)

Resetting the trips

On the Trip A screen, a short press of the upper CLR resets Trip A to 0, and a short press of the lower CLR resets Trip B. Trip B can also be reset on the clock, speed and final time screens and during a stage.

Setting any distance and measuring from it

1. On the Trip A screen, long press the upper CLR to enter Trip A.
2. Upper CLR adds 1 to the blinking digit, lower CLR subtracts 1. The blue button moves to the next digit.
3. Long press the upper CLR to confirm.
4. For Trip B, long press the lower CLR on the Trip A screen. Confirm with a long press of the lower CLR as well.



Before



Set to 12.50

Direction switch

Switch	Action	White LED
↑ Forward	Counts distance.	Off
○ Stop	Does not count distance.	On
↓ Reverse	Subtracts the distance driven (e.g. when going back after a wrong turn).	Blinking

It works on both Trip A and Trip B.

CHAPTER 06

Distance calibration

When the displayed distance does not match the road book, adjust it with the calibration factor.

Example: a section of 10.00 km in the road book shows 10.20 km on Trip A.

1. Long press the red button to enter calibration (C).
2. Upper CLR adds 1 to the blinking digit, lower CLR subtracts 1. The blue button moves to the next digit.
3. Set 10.20 ÷ 10.00 = 1.0200.
4. A short press of the red button saves and returns to the clock. Trip A now shows 10.00 km.

How to calculate
Calibration factor = ARM displayed distance ÷ road book distance



Calibration screen



Set to 1.0200

A long press of the red button during calibration returns the factor to 1.0000. The factor also applies to the speed display.

CHAPTER 07

Speed and maximum speed

The speed screen shows “S” followed by the calibrated speed. A short press of the upper CLR shows the maximum speed (P); press again to return. The maximum speed is recorded during stage timing and resets to 0 when a stage starts.



Speed



Maximum speed

CHAPTER 08

Speed alert

1. A short press of the 60 button opens the speed alert setting (A).
2. Upper CLR adds 10, lower CLR subtracts 10 (10–90 km/h).
3. A short press of the 60 button turns the alert ON and returns to the previous screen.
4. To turn it OFF, press the 60 button briefly again.

Speed	Red LED / “A” on the driver display (and the speed on the 2-row version)
Up to 5 km/h below the limit	On
From 5 km/h below the limit	Blinks slowly
At or above the limit	Blinks fast

A long press of the 60 button also opens the setting (leaving with a long press does not turn the alert ON).

CHAPTER 09

Stage timing and stage records

Stage timing

1. A short press of the red button starts a countdown to 00 s of the next minute. Trip A and Trip B are reset to 0 automatically.
2. If the car starts moving during the countdown (5 km/h or more), the stage starts at that moment. If it does not move, the stage starts at 00 s. You can also press the red button during the countdown to start immediately.
3. For 5 seconds after the start the time is shown, then the upper display shows Trip A. The blinking “SS” on the left shows that timing is running.
4. Every 1 km, the passing time blinks for 3 seconds.
5. Press the red button at the finish. The time blinks and Trip B returns to 0.
6. Press the red button again to end the stage. The clock screen shows the finish time blinking for 2 minutes (the upper CLR clears it at once).



Countdown



Running (Trip A)



Finish

The blue button cancels during the countdown. The red button on the remote and the external switch (yellow wire) do the same as the red button.

Stage records

Switch the upper display to “stage records” to view the recorded times. The lower display shows the record number (SS01–SS30). Upper CLR shows the next record, lower CLR the previous one. Up to 30 records are kept; they are lost when the power is off.

CHAPTER 10

Ave and final time

When an Ave (target average speed) is set, the final time shows the difference from driving at the Ave, based on the distance and elapsed time since the stage start. When the Ave is OFF, “OFF” is shown.

This function gives a guide to the time remaining until your TC-in when you arrive at the next time control (TC). It helps you check how much time you have in hand.

How to set

1. On the final time screen, a short press of the upper CLR opens the Ave setting.
2. Set the Ave on the lower display with the upper CLR: OFF → 30 → 40 ... The blue button moves to the next digit.
3. The upper display shows the start time. It is set automatically to the start time when a stage starts.
4. Long press the upper CLR to save (the distance so far is kept). Saving with the red button restarts the distance from that point.



21 s ahead



1 min 5 s behind

Blue LED warning
The blue LED (above the white LED) shows the final time margin: it blinks fast when the margin is under 5 minutes (including when behind), slowly from 5 to 10 minutes, and is off otherwise. It works once an Ave is set and a stage has been started (off while the stage screen is shown).

CHAPTER 11

Settings (time offset, brightness, driver display)

Long press the blue button to enter the settings; each short press of the blue button moves to the next setting. The last press returns to the previous screen.

Order	Display	Setting	Operation
1	Utc	Time offset from UTC (−12 to +12), e.g. UK 0, France +1 (winter), Japan +9.	Upper CLR: +1 Lower CLR: −1
2	LEd	Brightness (3 levels, 0 to 2).	Upper CLR: brighter Lower CLR: dimmer
3	dd	Driver display distance format. “0.01” is normal (12.34 km); “10” enlarges distances under 10 km to 1 m (1.230 km).	Upper / lower CLR: switch



Time offset



Brightness



Distance format

The settings are kept when the power is off.

About the driver display distance format

The “10” mode is mainly for recce: it lets the driver judge the length of straights and corners in 10 m steps. Seeing 10 m, 20 m … makes it easier to write clear pace notes.

CHAPTER 12

Remote control

Connect a remote control to the remote socket on the right side. The remote does not work while a setting is open.

Supplied remote (1 button)

Button	Function
White	Resets Trip B (normal screens and the stage screen)

Mount it near the steering wheel or gear lever to reset Trip B at each road book box without reaching for the unit.

5-button remote (optional)

A remote that also adjusts Trip A, switches the display and operates the stage timing.

Button	Function
Gray	Resets Trip B
Green	Trip A +10 m
Orange	Trip A −10 m
Blue	Switches the upper display (same as a short press of the blue button)
Red	Stage (same as a short press of the red button)

CHAPTER 13

Driver display (optional)

The driver display is an optional accessory.

Connect a driver display to the rear connector to show information in front of the driver. Two versions are available: the **2-row version** and the **1-row version**.

	2-row version	1-row version
Speed	Upper row (km/h)	Not shown
Distance	Lower row: speed alert “A” on the left, then Trip B	Speed alert “A” on the left, then Trip B
Speed alert	“A” and the speed blink	“A” blinks



2-row version



1-row version (distance only)



2-row (under 10 km enlarged)

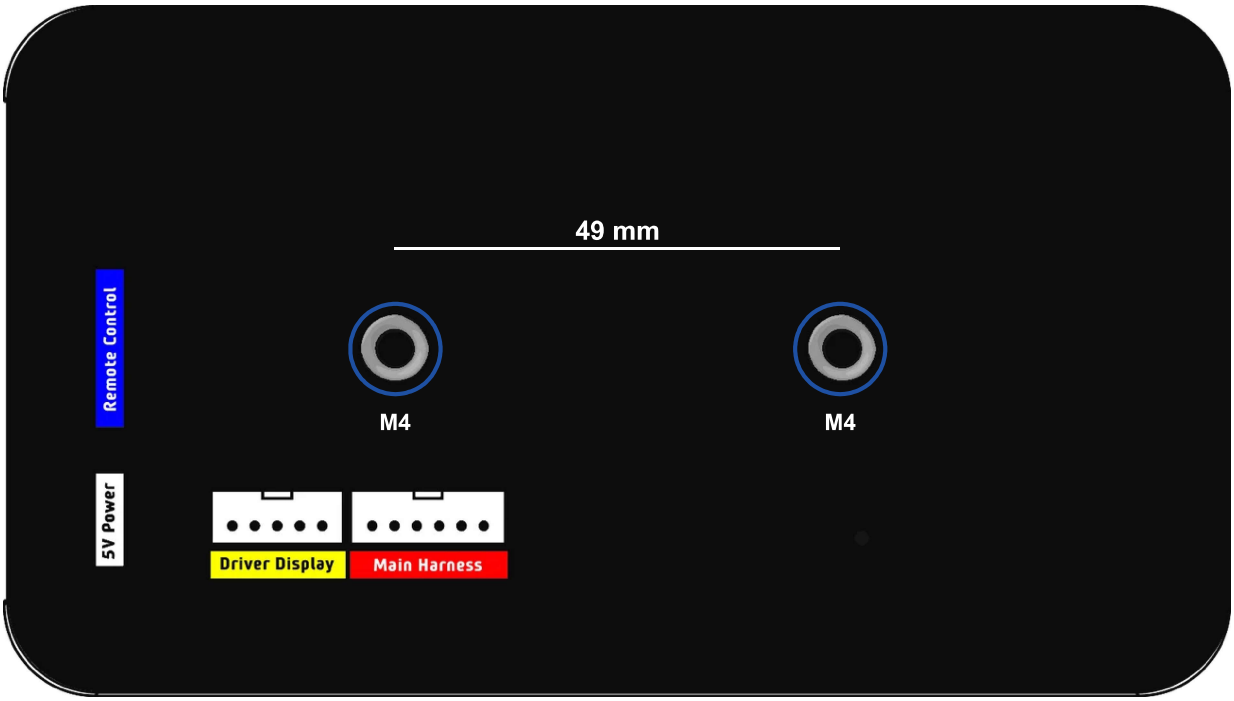


1-row (under 10 km enlarged)

Trip B and speed are calibrated values, the same as on the main unit. The distance format (normal / under 10 km enlarged) is selected in the settings (chapter 11) for both versions.

CHAPTER 14

Rear panel (mounting, connectors, external switches)



Mounting insert nuts

Two mounting nuts are embedded above the rear connectors, so the unit can be bolted to a bracket or mount.

Item	Details
Bolt	M4
Bolt length	Up to the thickness of the mounting part (bracket etc.) + 10 mm
Nut spacing	49 mm (centre to centre). RAM Mount bases with the same spacing can be used.

▲ Caution
A bolt that is too long may damage the inside of the unit. Always use bolts no longer than “thickness + 10 mm”.

Connectors

There are two connectors on the rear: the left one is for the driver display, the right one for the main harness.

Main harness pin assignment

Pin (from the left, seen from the rear)	Color	Use
1	Red	DC12V
2	Black	GND
3	White	Reset trip (external switch: Trip B reset)
4	Yellow	Stage (external switch: stage)
5	Black	GND
6	Black	GND

External switches

Wire	Function
White (Reset trip)	Trip B reset only. A short press resets Trip B on the normal screens and the stage screen (not while setting). Useful for adding a foot switch.
Yellow (Stage)	Adds only the red button's stage function (start / finish). A long press does nothing (it does not open calibration).

Wiring
Connect one side of a momentary switch (ON only while pressed) to the white or yellow wire and the other side to the vehicle GND. If the vehicle GND is not available, use one of the two GND (black) wires on the right of the connector.

CHAPTER 15

Specifications

Item	Details
Dimensions	W 135 × H 76 × D 27 mm (excluding buttons, switches and other protrusions)
Power	USB-C (5 V) or DC12V (main harness on the rear). Do not use both at the same time.
Distance / speed	Measured by GPS. Calibration factor 0.0000–2.9999
Memory	Trip A, Trip B, calibration, time offset, brightness, driver display distance format, speed limit and Ave settings (stage records are lost when the power is off)
Connections	Driver display, main harness (DC12V, external switches), remote control
Supplied accessory	Remote control (white, 1 button: Trip B reset)
Options	5-button remote control, driver display (2-row / 1-row version)

CHAPTER 16

Quick reference

Screen	Blue	Red	Upper CLR	Lower CLR
Clock	Short: next screen Long: settings	Short: stage Long: calibration	Clears the blinking finish time	Trip B reset
Trip A	Short: next screen	Short: stage Long: calibration	Short: Trip A reset Long: enter Trip A	Short: Trip B reset Long: enter Trip B
Speed	Short: next screen	Short: stage Long: calibration	Speed ⇔ maximum	Trip B reset
Final time	Short: next screen	Short: stage Long: calibration	Short: Ave setting	Trip B reset
Stage records	Short: to clock	Short: stage	Next record	Previous record
Stage	Countdown: cancel	Countdown → start → finish → end	After finish: end	Trip B reset

Setting screen	Upper CLR	Lower CLR	Blue	Confirm / back
Calibration (C)	+1	−1	Next digit	Red short: save / Red long: 1.0000
Enter Trip A / B	+1	−1	Next digit	A: upper CLR long / B: lower CLR long
Utc / LEd / dd	+ / switch	− / switch	Next setting	Last blue press returns
Speed alert (A)	+10	−10	—	60 short: back with alert ON
Ave setting	+	−	Next digit	Upper CLR long: save / Red: save and restart the distance

An animated manual is also available: manual.rally-arm.com